

NSS COLLEGE OF ENGINEERING PALAKKAD – 678008

**SIMULATION AND ANALYSIS OF VIBRATION BASED
ENERGY HARVESTING**

PROJECT REPORT

Submitted in partial fulfilment of the requirements for EEL334 Power Electronics Lab

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ABSTRACT

The growing demand for sustainable and autonomous power sources for low-power electronic systems has intensified interest in energy harvesting techniques. This project presents the design and MATLAB/Simulink-based simulation of a vibration-driven energy harvesting system capable of converting mechanical vibrational energy into regulated electrical power. A sinusoidal waveform is used to model ambient vibrational inputs, representing real-world sources such as machinery vibrations and structural oscillations. The input signal is applied to an AC electrical equivalent circuit and processed through a full-wave rectifier to obtain a unidirectional pulsating DC output. A passive filtering stage is incorporated to minimize ripple and produce a smoother DC voltage. This filtered output is then supplied to a DC-DC buck converter, which regulates and steps down the voltage according to load requirements using high-frequency switching and pulse width modulation (PWM) control. The system is analyzed under steady-state conditions, evaluating key performance parameters including output voltage, ripple factor, and conversion efficiency. Simulation results demonstrate that the proposed system effectively converts low-amplitude vibrational energy into a stable DC supply suitable for powering low-power electronic devices. The study confirms the feasibility of vibration-based energy harvesting systems and highlights their potential applications in self-powered sensors, wireless monitoring systems, and embedded electronics, contributing to the advancement of sustainable and maintenance-free energy solutions.

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INTRODUCTION

In the modern era of smart systems, miniaturized electronics, and advanced wireless communication technologies, the demand for reliable, continuous, and sustainable power sources has increased significantly. With the rapid growth of portable electronic devices, wireless sensor networks, and Internet of Things (IoT) applications, conventional batteries are becoming less suitable due to their inherent limitations such as finite lifespan, frequent replacement, increased maintenance costs, and environmental hazards associated with disposal. These drawbacks have driven the need for alternative energy solutions that are both efficient and environmentally friendly. Energy harvesting has emerged as a promising technique to address these challenges by capturing small amounts of energy available in the surrounding environment and converting it into usable electrical power. Among various ambient energy sources such as solar, thermal, and electromagnetic energy, mechanical vibrations are particularly attractive due to their widespread availability in both natural and industrial settings. Vibrations are commonly generated by machinery, vehicles, rotating equipment, and structural systems such as bridges and buildings. These vibrations, which are usually wasted, can be effectively harnessed and converted into electrical energy, thereby improving overall energy efficiency and enabling the development of self-sustaining electronic systems. The vibration-based energy generator is designed to utilize this ambient mechanical energy and convert it into electrical power suitable for low-power applications. The core component of this system is a piezoelectric transducer, which operates on the principle of the piezoelectric effect, where certain materials such as quartz and lead zirconate titanate (PZT) generate an electrical charge when subjected to mechanical stress or strain. When exposed to continuous vibrations, these materials produce an alternating electrical output corresponding to the applied mechanical force. However, the generated output is typically low in magnitude and irregular, making it unsuitable for direct use in electronic devices, which require a stable direct current (DC) supply.

To make the harvested energy usable, it undergoes several stages of power conditioning and regulation. Initially, the alternating current (AC) signal produced by the piezoelectric transducer is converted into a unidirectional pulsating DC signal using a full-wave rectifier, ensuring that both halves of the waveform are utilized for improved efficiency. This is followed by a filtering stage using passive components such as capacitors to reduce ripple content and produce a smoother and more stable DC output. The filtered voltage is then fed into a DC-DC buck converter, which regulates and steps down the voltage to a desired level based on load requirements. The buck converter operates using high-frequency switching techniques and employs components such as inductors, diodes, capacitors, and controlled switches. Pulse Width Modulation (PWM) is used to control the duty cycle, enabling precise voltage regulation and efficient energy transfer. Additionally, energy storage elements such as capacitors are incorporated to store excess energy and provide a continuous power supply during fluctuations in vibration levels. The entire system is modeled and simulated using MATLAB/Simulink, allowing detailed analysis of waveform characteristics and performance parameters such as output voltage, current, ripple factor, and conversion efficiency under steady-state conditions. This project aims to demonstrate an effective method of converting ambient vibrational energy into usable electrical power, reducing dependence on conventional batteries and enabling maintenance-free operation. The proposed system has wide-ranging applications in wireless sensor networks, structural health monitoring, wearable devices, industrial automation, and IoT systems, where long-term, reliable, and autonomous power sources are essential. Thus, vibration-based energy harvesting represents a sustainable and innovative solution for powering next-generation electronic systems while contributing to energy efficiency and environmental conservation.

OBJECTIVE

1. To design and simulate a vibration-based energy harvesting system using a vibrational source, where the mechanical vibrations are modeled as a sinusoidal input and converted into electrical energy through rectification and filtering stages.
2. To obtain and analyze key performance graphs, including output voltage, output current, power characteristics, input voltage, and rectified input, in order to evaluate the efficiency and behavior of the energy harvesting system.

DESIGN AND OPERATIONAL ANALYSIS

A vibration based energy harvester converts mechanical vibrations from the surroundings into electrical energy. In this circuit, the vibration source produces an AC voltage which is converted into DC using a rectifier circuit made of diodes. The capacitor filters the rectified output to reduce ripple, while the inductor and diode help regulate and store the energy. The filtered DC power is then supplied to the load, demonstrating how vibrational energy can be harvested and used to power small electronic devices.

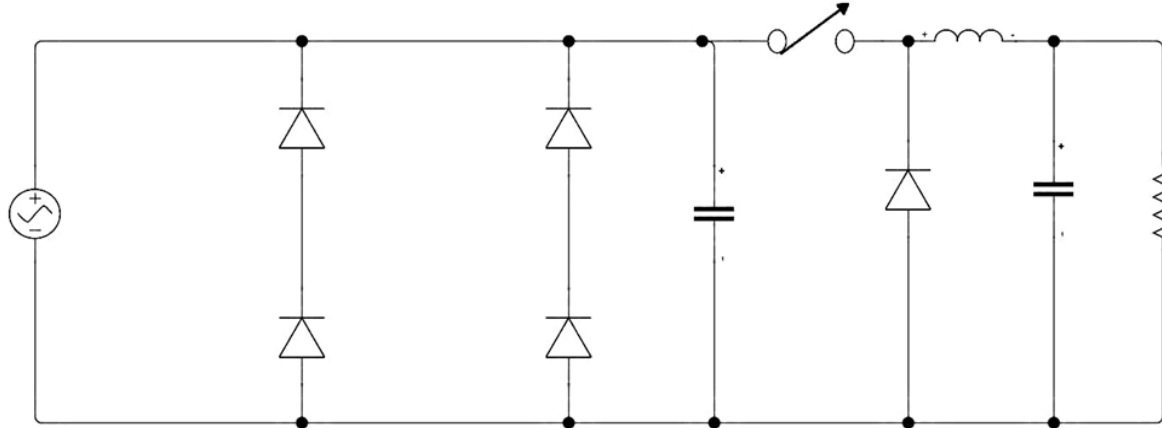


Fig 1. Circuit Diagram of Vibration based energy harvester

Operational Analysis

The operation of a buck converter is studied by analyzing its switching behavior and the energy transfer between its components. Important performance parameters include output voltage, inductor current ripple, and efficiency. The output voltage of the converter is controlled by varying the duty cycle of the switching signal. Changes in input voltage or load conditions influence the converter's performance, and in open-loop operation, the output voltage is directly proportional to the duty cycle.

The buck converter operates using an inductor, capacitor, diode, and a controlled switch (typically a MOSFET). Its operation can be explained in two modes.

- Switch ON state

When the switch is turned ON, the input voltage is directly applied across the inductor and load. The inductor starts storing energy, causing its current to increase linearly. During this period, the diode becomes reverse-biased and does not conduct. The capacitor charges and helps in maintaining a smooth output voltage while supplying the load.

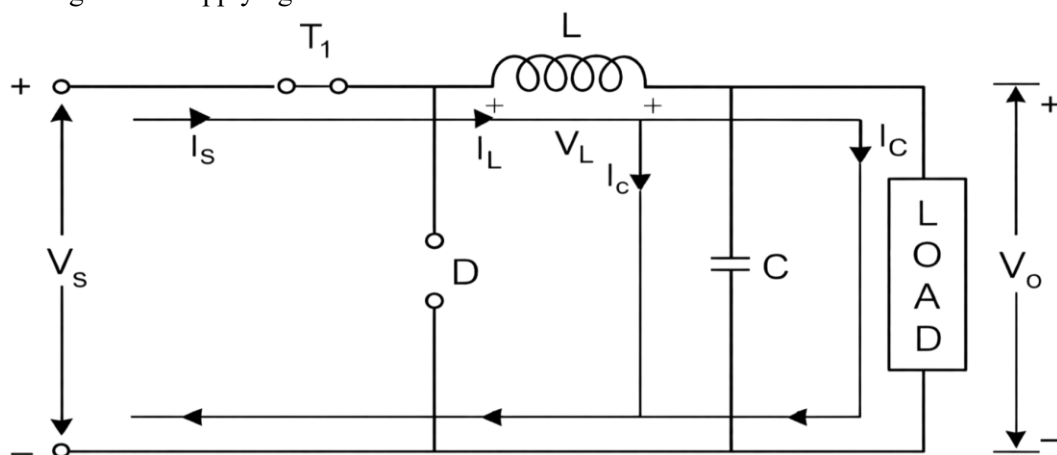


Fig1.1 : Diagram of buck converter when switch is ON

- Switch OFF state

When the switch is turned OFF, the inductor releases its stored energy to the load through the diode. The diode becomes forward-biased, allowing current to flow. The inductor current decreases gradually, but continues to supply the load along with the capacitor, thereby maintaining a continuous and regulated output voltage

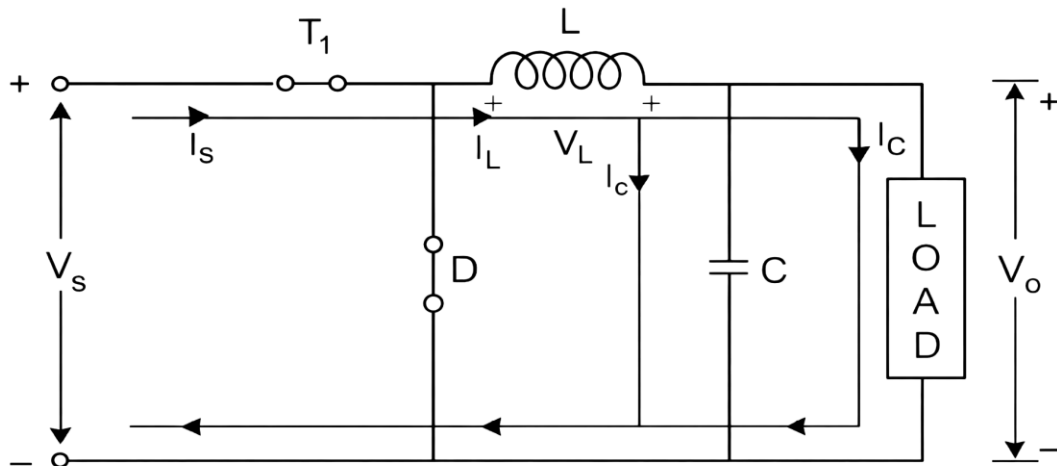


Fig1.2 : Circuit Diagram of buck converter when switch is OFF

Apply KVL in fig 1.2

$$V_L = V_{in} - V_o$$

$$V_L = L \frac{di}{dt}$$

So,

$$L \frac{di}{dt} = V_{in} - V_o$$

$$\frac{di}{dt} = \frac{V_{in} - V_o}{L}$$

Change in inductor current during ON time:

$$\Delta I_L(\text{ON}) = \frac{V_{in} - V_o \cdot T_{on}}{L}$$

Apply KVL in Fig 1.2

$$T_{on} V_L = -V_o$$

$$L \frac{di}{dt} = -V_o$$

$$\frac{di}{dt} = \frac{-V_o}{L}$$

Change in current:

$$\Delta I_L(\text{OFF}) = \frac{V_o \cdot T_{off}}{L}$$

For periodic operation:

$$\Delta I_L(\text{ON}) = \Delta I_L(\text{OFF})$$

$$D = \frac{T_{on}}{T}$$

$$\Delta I_L = \frac{(V_{in} - V_o) \cdot D \cdot T}{L}$$

Since

$$T = \frac{1}{f}$$

$$\Delta I_L = \frac{(V_{in} - V_o) \cdot D}{L \cdot f}$$

Inductance Design Formula

$$L = \frac{(V_{in} - V_o) \cdot D}{\Delta I_L \cdot f}$$

Charge change

$$\Delta Q = C \cdot \Delta V_o$$

Also,

$$\Delta Q = \frac{\Delta I_L \cdot T}{2} \cdot \frac{T}{2}$$

$$\Delta Q = \frac{\Delta I_L \cdot T}{8}$$

Equating

$$C \cdot \Delta V_o = \frac{\Delta I_L \cdot T}{8}$$

$$C = \frac{\Delta I_L \cdot T}{8 \cdot \Delta V_o}$$

Substitute

$$T = \frac{1}{f}$$

Capacitance Design Formula

$$C = \frac{\Delta I_L}{8 \cdot f \cdot \Delta V_o}$$

ANALYSIS OF SIMULATION RESULT

The simulation results confirm that the proposed vibration-based energy harvesting system effectively converts low-amplitude mechanical vibrations into a stable DC output suitable for low-power applications. The AC input, modeled as a sinusoidal vibration source, is successfully rectified using a full-wave rectifier, producing a pulsating DC signal which is then smoothed by the passive filtering stage, significantly reducing ripple content. The DC–DC buck converter, controlled using high-frequency PWM switching, efficiently regulates the output voltage to meet load requirements under steady-state conditions. Performance evaluation indicates satisfactory voltage regulation, low ripple factor, and improved conversion efficiency, demonstrating reliable operation of the system. Overall, the results validate the feasibility of the design for powering small-scale electronic devices and highlight its potential for use in self-powered and maintenance-free applications such as wireless sensors and embedded systems.

Simulation Parameters

PARAMETERS	VALUES
INPUT VOLTAGE	20 V
OUTPUT VOLTAGE	8.5 V
INDUCTANCE	$1.7 \times 10^{-3}\text{mH}$
CAPACITANCE	$5.88 \times 10^{-6}\text{ F}$
RESISTANCE	$8.5\ \Omega$
FREQUENCY	25 KHZ
DUTY RATIO	42.5%

Table 1 : Simulation Parameters

The Fig 2 represents a sine wave, which is a vibrational source alternating between positive and negative cycles. This is the raw AC supply given as a sinusoidal waveform before any rectification process. It shows the continuous oscillation of voltage over time, forming the basis for conversion into DC in the rectifier circuit.

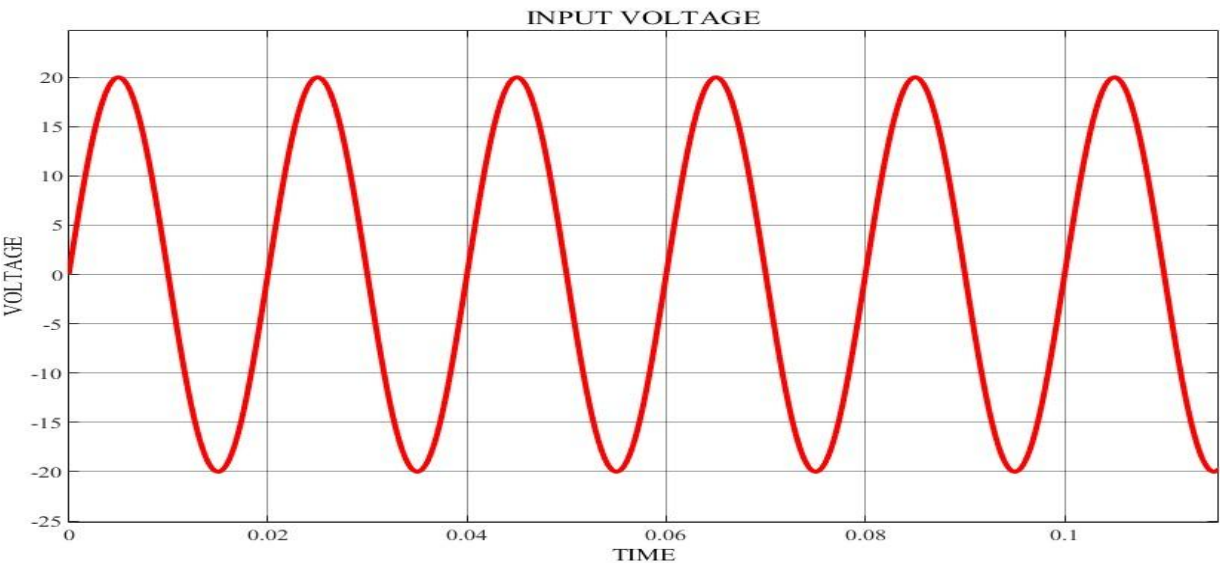


Fig 2 : Simulation graph of Input Voltage

In Fig 3, it shows how the alternating sine wave input is converted into a unidirectional signal. Instead of oscillating above and below zero like the AC input, the waveform now stays on the positive side, representing direct current. However, it is not perfectly smooth — small ripples remain due to incomplete filtering. This demonstrates the rectifier's role in transforming AC into DC, which can then be further refined using filters to achieve a more stable voltage supply.

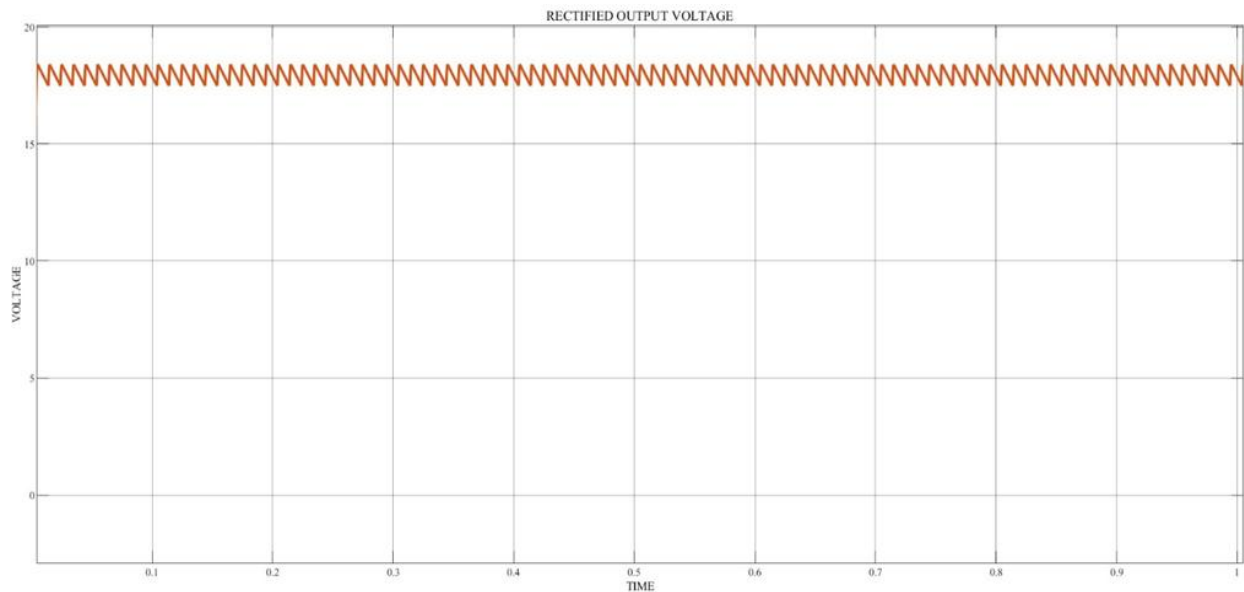


Fig 3 : Simulation graph of Rectified input voltage

$$\begin{aligned} \text{Ripple} &= V_{\text{OUT}} / V_{\text{IN}} \\ &= 0.8 / 20 \\ &= 0.425 \end{aligned}$$

In Fig 4, it shows the current delivered to the load. It follows the rectified voltage pattern, meaning the current is steady but still has minor fluctuations. The smoother the voltage, the steadier the current supplied to the circuit.

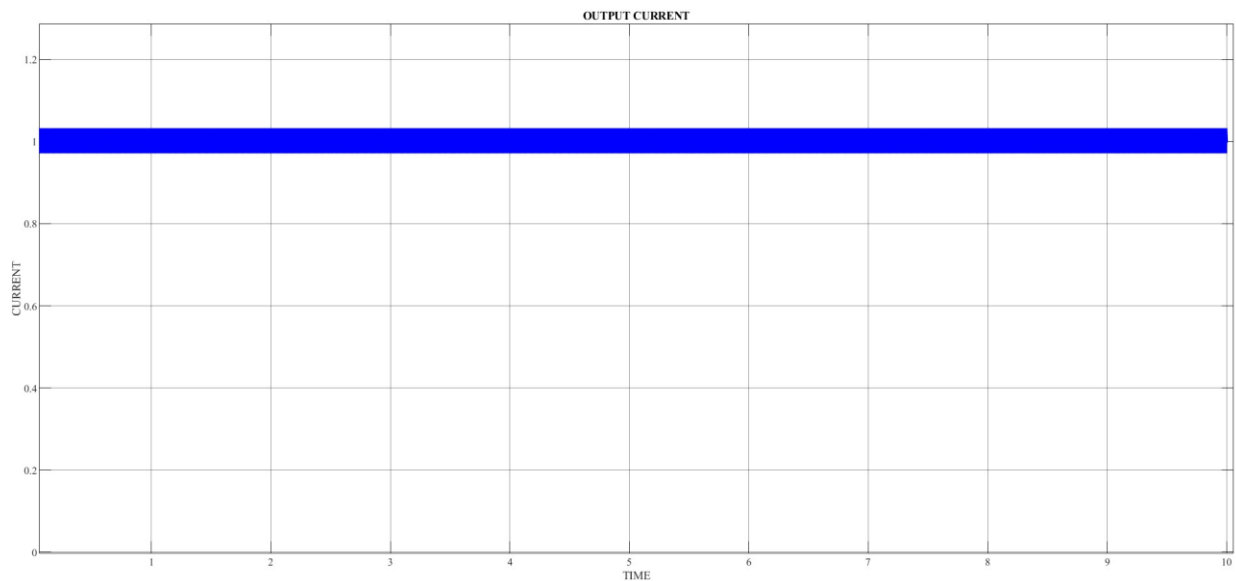


Fig 4 : Simulation graph of output current

In Fig5, the output voltage graph shows the steady DC voltage that appears after rectification and filtering. It remains almost constant over time, unlike the alternating input, and represents the usable supply delivered to the load. This graph highlights the stability of the voltage, which is essential for powering electronic circuits without interruption. In short, it demonstrates how the rectifier transforms the input into a reliable and continuous DC output.

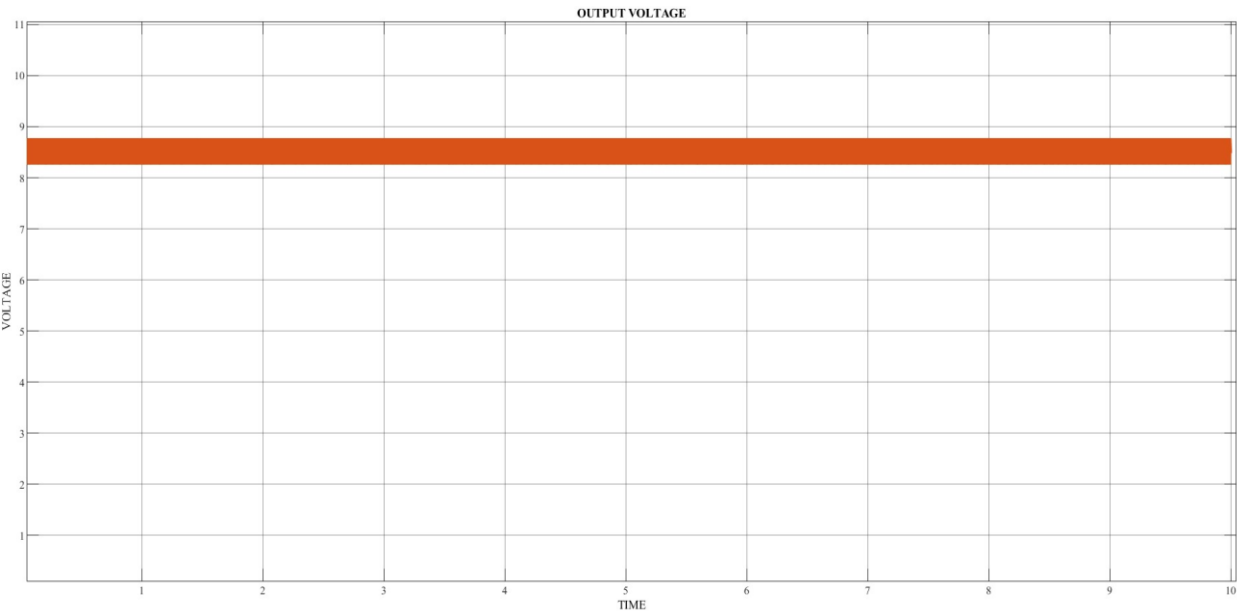


Fig 5 : Simulation graph of output voltage

Fig 6 plots voltage against current, showing how the circuit behaves under different load conditions. It helps in analyzing efficiency, power transfer, and the relationship between input and output. This is used to evaluate how effectively the rectifier delivers usable power.

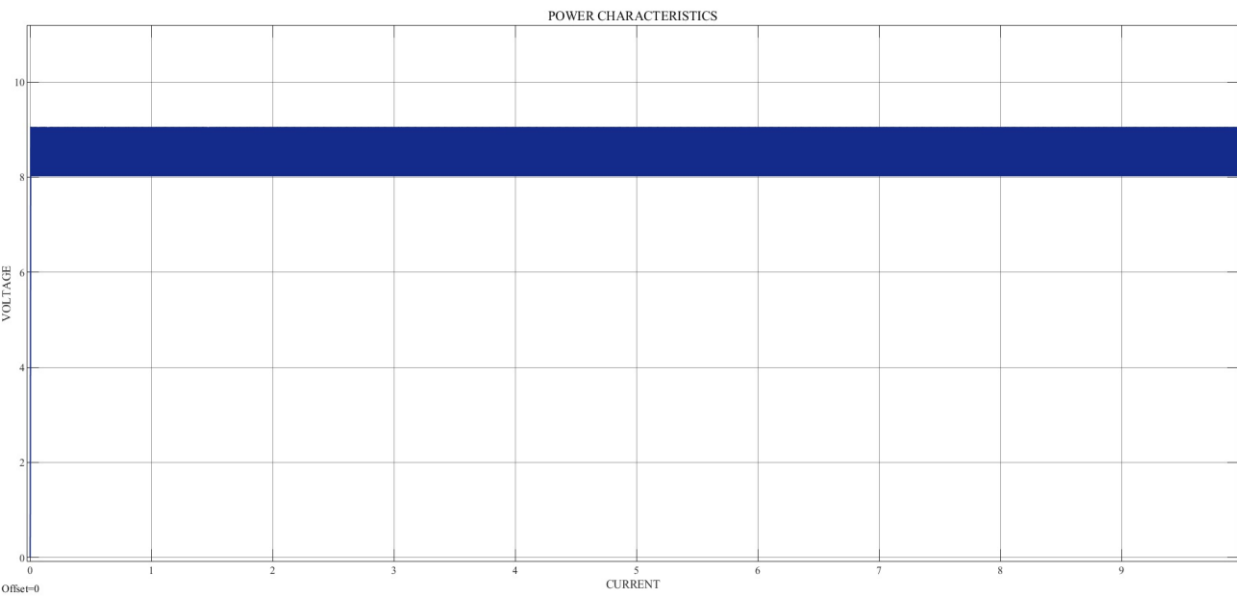


Fig 6 : Simulation graph of Power character

CONCLUSION AND FUTURE SCOPE

Conclusion:

The vibration-based energy harvesting system developed in this project successfully demonstrates the feasibility of converting ambient mechanical vibrations into a usable and regulated electrical output. By modeling real-world vibrational sources as a sinusoidal input in MATLAB/Simulink, the system effectively captures and processes low-amplitude energy through successive stages including rectification, filtering, and DC-DC voltage regulation. The integration of a full-wave rectifier ensures efficient AC-to-DC conversion, while the passive filter significantly reduces output ripple, improving the quality of the DC signal. Furthermore, the implementation of a buck converter with PWM control provides reliable voltage regulation, making the output suitable for practical low-power applications. The steady-state analysis confirms acceptable performance in terms of output voltage stability, ripple reduction, and conversion efficiency. Overall, this project validates that vibration energy harvesting is a promising and sustainable solution for powering small-scale electronic systems, especially in environments where conventional power sources are limited or impractical.

Future

Scope:

The scope for further development of this vibration energy harvesting system is substantial, with opportunities to enhance both efficiency and real-world applicability. Future work can focus on incorporating advanced transducers such as piezoelectric, electromagnetic, or electrostatic devices to improve energy conversion efficiency under varying vibration conditions. The system can be extended to operate under dynamic and non-linear vibration profiles rather than ideal sinusoidal inputs, making it more adaptable to real-world environments. Additionally, integrating energy storage elements such as supercapacitors or rechargeable batteries can enable continuous power supply even during periods of low vibration. Optimization of the DC-DC converter using modern control techniques, such as adaptive or AI-based control, can further enhance performance and efficiency. Hardware implementation and real-time testing will be a crucial next step to validate simulation results and assess practical challenges. Moreover, the system can be scaled and customized for applications in structural health monitoring, industrial IoT, wearable electronics, and smart infrastructure, contributing significantly to the advancement of self-powered and maintenance-free electronic systems.

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